

JEE MAIN REVISION MASTER: INDEFINITE INTEGRALS

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 18 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Standard Integrals, Substitution & By Parts	17 PYQs	Integration Techniques
Standard	Partial Fractions & Special Algebraic Integrals	1 PYQs	Algebraic Integrals

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Standard Integrals, Substitution & By Parts	$\int e^x [f(x) + f'(x)] dx = e^x f(x) + C$, $\int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \ln \left \frac{x-a}{x+a} \right + C$, $\int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1} \left(\frac{x}{a} \right) + C$
Partial Fractions & Special Algebraic Integrals	$\int \frac{dx}{x^2 + a^2} = \frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right) + C$, $\int \frac{dx}{\sqrt{x^2 \pm a^2}} = \ln x + \sqrt{x^2 \pm a^2} + C$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Standard Integrals, Substitution & By Parts (Very High Yield — 17 PYQs)

Core Principle: Fundamental integrals, substitution methods, ILATE integration by parts, and linear/quadratic partial fractions.

■ **Caution / JEE Pitfall:** In integration by parts, follow the ILATE priority strictly for choosing the first function u (Inverse, Logarithmic, Algebraic, Trigonometric, Exponential).

■ **Speed Trick:** The pattern $\int e^x [f(x) + f'(x)] dx = e^x f(x) + C$ appears with high frequency in both JEE Main and KCET.

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (Numerical): "If $\int (\sin x)^{\frac{-11}{2}} (\cos x)^{\frac{-5}{2}} dx = -\frac{1}{p_1 q_1} (\cot x)^{\frac{1}{q_2}} - \frac{1}{p_2 q_2} (\cot x)^{\frac{5}{2}} - \dots$ " [Ans: 16]
- [2025 Evening Shift] Q#3 (Numerical): "If $\int \frac{(\sqrt{1+x^2}+x)^{10} (\sqrt{1+x^2}-x)^9 dx}{m((\sqrt{1+x^2}+x)^n (n\sqrt{1+x^2}-x))+C}$ where C is the c..." [Ans: 379]
- [2025 Evening Shift] Q#4 (Numerical): "If $\int \frac{2x^2+5x+9\sqrt{x^2}+x+1}{x^2+x+1+\alpha\sqrt{x^2}+x+1+\beta \log_e(x)} dx = \sqrt{x^2}+x+1+C, \dots$ " [Ans: 16]

3.2 Partial Fractions & Special Algebraic Integrals (Standard Yield — 1 PYQs)

Core Principle: Integration using partial fraction decompositions, linear over quadratic forms, and standard quadratic root substitutions.

■ **Caution / JEE Pitfall:** Check if degree of numerator $\geq$ degree of denominator; if so, polynomial long division **MUST** be performed first.

■ **Speed Trick:** For integrals of type $\int (dx / a + b \cos^2 x)$, divide numerator and denominator by $\cos^2 x$ and substitute $\tan x = t$.

Representative Source PYQs:

• **[2025 Evening Shift] Q#2 (Numerical):** "If $\int \left(\frac{1}{x} + \frac{1}{x^3} \right) (\sqrt{23}) dx = -\frac{\alpha}{3(\alpha+1)} (3x^\gamma \beta + x^\gamma)^\alpha + C, x > 0, (\alpha, \beta, \gamma \in \mathbb{Z})$ " [Ans: 19]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 22nd January Morning Shift (Q#1)

If $\int (\sin x)^{-11/2} (\cos x)^{-5/2} dx = -\frac{p_1 q_1}{p_2 q_2} (\cot x)^{9/2} - \frac{p_2 q_2}{p_3 q_3} (\cot x)^{5/2} - \frac{p_3 q_3}{p_4 q_4} (\cot x)^{1/2} + \frac{p_4 q_4}{p_5 q_5} (\cot x)^{-3/2} + C$, where p_i and q_i are positive integers with $\gcd(p_i, q_i) = 1$ for $i = 1, 2, 3, 4$ and C is the constant of integration, then $\frac{15p_1 p_2 p_3 p_4 q_1 q_2 q_3 q_4}{15}$ is equal to ____

Correct Answer: 16

Solution Analysis: $\int (\sin x)^{-11/2} (\cos x)^{-5/2} dx = \int \frac{1}{\tan^{11/2} x} \cdot \sec^8 x dx = \int \frac{(1 + \tan^2 x)^3 \sec^2 x \tan^{11/2} x dx}{\tan^{11/2} x}$
 put $\tan x = t$ $\frac{15p_1 p_2 p_3 p_4 q_1 q_2 q_3 q_4}{15} = (15(2 \times 6 \times 6 \times 2) / 9 \times 5 \times 1 \times 3) = 16$

Benchmark Problem 2: JEE Main 2025 (Online) 7th April Evening Shift (Q#2)

If $\int \left(\frac{1}{x} + \frac{1}{x^3} \right) (\sqrt{23}) dx = -\frac{\alpha}{3(\alpha+1)} (3x^\beta + x^\gamma)^{\alpha+1/\alpha} + C, x > 0, (\alpha, \beta, \gamma \in \mathbb{Z})$, where C is the constant of integration, then $\alpha + \beta + \gamma$ is equal to _____.

Correct Answer: 19

Solution Analysis: $\int \left(\frac{1}{x} + \frac{1}{x^3} \right) (\sqrt{23}) dx$ using $t = (3/x) + \frac{1}{x^3} \Rightarrow dt = -3 \left(\frac{1}{x^2} + \frac{1}{x^4} \right) dx$
 $\int \frac{1}{\sqrt{23}} dt = -\frac{1}{\sqrt{23}} \ln |t| + C = -\frac{1}{\sqrt{23}} \ln \left| \frac{3}{x} + \frac{1}{x^3} \right| + C \Rightarrow \alpha = 23, \beta = -1, \gamma = -3 \Rightarrow \alpha + \beta + \gamma = 19$